

Jayson C. Sepeda

sepedajayson01@gmail.com | 09303865907 | [Iie Sepeda | Portfolio](#) | Manila, Philippines

Computer Science student graduating October 2025 from Pamantasan ng Lungsod ng Maynila, on track to graduate Magna Cum Laude, with hands-on experience in application development, system design, and programming across multiple languages (**Java, C, C++, Python, PHP, Dart**). Skilled in delivering full-cycle software projects, troubleshooting system issues, and optimizing workflows. Recognized internationally for research in machine learning and speech emotion recognition. **Quick to adapt to new tech stacks including enterprise platforms.**

Education

Pamantasan ng Lungsod Ng Maynila

2021-2025

Bachelor of Science in Computer Science | GWA: 1.44 | Magna Cum Laude

Core Skills

- **Programming:** Java, C, C++, Python, PHP, Dart, HTML, CSS, JavaScript, OOP, Data Structure and Algorithms.
- **Application/Software Development:** React, Django, Laravel, Flask, Flutter, AppSheet, MySQL
- **Enterprise Tools:** Looker Studio, Git/GitHub, Google Apps Script, UNIX Shell (basic)
- **Soft Skills:** Problem-solving, Analytical Thinking, Team Collaboration, Time Management, Quick Learner.

Key Achievements & Published Works

- **Best Abstract Award Winner** - 1st ASEAN Multidisciplinary Research Conference 2024 (International Society of Transdisciplinary Researchers, Educators, and Leaders Inc.)
- **Published Author** - *Enhancement of Gated Recurrent Unit for Speech Emotion Recognition in the Implementation of Voice-Based Danger Recognition System* (Published at United International Journal for Research & Technology, Indexed on Google Scholar and ResearchGate)
 - DOI: <http://dx.doi.org/10.13140/RG.2.2.21381.26085>

Experience

Application Developer & IT Intern | Third Generation Holdings Corporation July 04, 2024 – August 23, 2024

- Created an internal document request, retrieval, and form creation application using Google AppSheet, significantly reducing manual labor and streamlining operational processes.
- Spearheaded end-to-end application testing by distributing a test version across all company departments to validate functionality, gather feedback, and ensure alignment with operational needs.
- Designed and deployed dynamic dashboards and reports using AppSheet and Looker Studio, improving internal reporting efficiency.
- Automated document generation workflows via Google Sheets and Apps Script, reducing manual work by ~50%.
- Customized forms and backend workflows to streamline data input and retrieval processes.
- Provided IT support for hardware/software troubleshooting and issue resolution across departments.
- Conducted system evaluations and submitted structured defect reports to improve internal tools.

Projects

Gated Recurrent Unit (GRU) Voice-Based Danger Recognition System | Lead Programmer

May 2024 – May 2025

Programming Language, Software & Framework: Python, Kaggle, Flask, React-Vite

- Built a GRU-based neural network to classify emotional states from speech signals (sad, happy, angry, fear, neutral, and disgust) using a custom Kaggle dataset.
- Identified and prepared the appropriate dataset, handling data preprocessing and formatting to ensure compatibility with the model pipeline.
- Improved model performance by addressing three core issues in the base GRU model and applying targeted enhancements to each.
- Created a full-stack web system with a React Vite frontend and Flask backend capable of accepting live microphone recordings or uploaded .wav files for emotion classification.
- Engineered backend logic to detect "fear" emotion and trigger a danger recognition response, enabling real-time alert capability.

- Designed and implemented the preprocessing pipeline, data splitting strategy, and model evaluation process to ensure training efficiency and reliable metrics.
- Authored the research manuscript and prepared the system for academic publication and thesis presentation.

PLM Archive System | Programmer and Documentation Lead

September 2024 – December 2024

Language & Framework: Python, Django, MySQL

- Developed core backend functionalities using Django and MySQL for a centralized web-based university thesis repository system.
- Implemented dynamic user roles and authentication logic (Admin, Faculty, and Student) to manage access, submission, and approval workflows.
- Built and styled responsive frontend pages using HTML, CSS, JavaScript, and Bootstrap, improving usability and navigation.
- Created reusable UI components and integrated backend APIs for seamless data handling and interaction.
- Used Git and GitHub for collaborative version control, managing feature branches and pull requests within a team environment.
- Co-authored system documentation, including a user manual, deployment guide, project brochure, and presentation materials.

Mobile Application Development (Renacimiento App & PLM App) | UI/UX & Lead Frontend Developer

March 2024 – June 2024

Software Development Kit & Programming Language: Flutter & Dart

- Led the UI/UX design process, creating detailed wireframes, low-fidelity, and high-fidelity designs in Figma.
- Spearheaded frontend development using Flutter and Dart, translating Figma designs into fully functional, responsive mobile interfaces.
- Facilitated regular team meetings and client check-ins to align on design direction, gather feedback, and track deliverables against deadlines.
- Developed modular and reusable UI components to ensure maintainability, design consistency, and ease of integration with backend services.
- Structured codebase for seamless collaboration with backend developers, enabling efficient full-stack integration.
- Participated in beta testing and QA, helping to identify, document, and resolve bugs and UI/UX inconsistencies prior to final release.
- Contributed to project documentation, including user manuals, promotional brochures, and system flow guides.

Licenses and Certifications

Artificial Intelligence Fundamentals

IBM SkillsBuild

Skills: Python Programming

July 2025

Python Essentials 1

CISCO Networking Academy and Python Institute

Skills: Python Programming

July 2025

Introduction to Cybersecurity

CISCO Networking Academy

Skills: Explain the most common Cyber Threats, Attacks, and Vulnerabilities.

March 2023

Cybersecurity Essentials

CISCO Networking Academy

Skills: Describe the procedures used by cybercriminals and explain the purpose of laws related to cybersecurity.

April 2023

Professional Reference

Bren Alfred España

Pamantasan ng Lungsod ng Maynila

- baespana24@gmail.com
- 09171388873
- Lecturer III / Professor
- Best time to call 3pm onwards